*A Project Report on*

# Virtual Try-On System

submitted in partial fulfilment of the requirements for the award of the degree of

## BACHELOR OF ENGINEERING

By


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**2025-2026**

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# CERTIFICATE

This is to certify that the project entitled **"Virtual T-Shirt Try-On System Using TypeScript, MediaPipe & OpenCV"** is a Bonafide work of **Vedang Bodas (24106192), Karan Borana(24106161), Nimish Bandal (24106075), Parini Gupta(24106180)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning)**.

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# Project Report Approval for S.E.

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# Abstract

The Virtual T-Shirt Try-On System is an augmented reality application that allows users to virtually try on T-shirts in real-time using a standard webcam or mobile camera. The system is primarily implemented in TypeScript and leverages the Media Pipe Pose library for precise human body landmark detection, along with OpenCV.js for image processing and augmented reality rendering within the browser environment. Pre-loaded T-shirt assets are mapped and overlaid onto the user's body based on detected skeletal key points, specifically the shoulder and torso landmarks, to simulate a realistic wearing experience without any physical contact.

The core motivation behind this project is to offer fashion brands—particularly new and emerging ones—a cost-effective, scalable, and engaging solution to enable virtual try-ons for their customers. The TypeScript codebase ensures type safety, maintainability, and seamless integration with modern web frameworks such as React and Next.js. By eliminating the need for physical trials, the system reduces return rates, improves customer satisfaction, and enhances purchase confidence in online shopping environments. The software is designed as a service platform (SaaS model) where brands can upload their garment catalogues and offer try-on experiences through web or mobile interfaces.

The project demonstrates real-time performance, accurate garment overlay, and ease of integration with e-commerce platforms. Future scope includes extending the system to support mobile augmented reality, 3D garment rendering, and AI-based size recommendation based on body measurements.

The SDG Goals are:

- **Climate impact of reverse logistics (SDG 13 – Climate Action)**
    - Transportation for deliveries and returns increases carbon footprint. Hence reducing it
- **Promoting sustainable consumption (SDG 12 – Responsible Consumption and Production)**
    - Overproduction and returned inventory contribute to textile waste.
- **Driving innovation in retail technology (SDG 9 – Industry, Innovation and Infrastructure)**
    - Encourages smart and sustainable technological solutions.

- **Reduce clothing return rates in online shopping** *(SDG 12 – Responsible Consumption and Production)*
- **Lower carbon emissions from logistics and product returns** *(SDG 13 – Climate Action)*
- **Enhance customer confidence through AI-powered virtual fitting** *(SDG 9 – Industry, Innovation and Infrastructure)*

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## ABBREVIATIONS

| | |
|---|---|
| AR | Augmented Reality |
| CV | Computer Vision |
| DOM | Document Object Model |
| FPS | Frames Per Second |
| ML | Machine Learning |
| SaaS | Software as a Service |
| SDK | Software Development Kit |
| TS | TypeScript |
| WASM | Web Assembly |

# Chapter 1
# Introduction

## 1. Introduction

The rapid growth of e-commerce and online fashion retail has transformed the way consumers discover and purchase clothing. However, one of the most persistent challenges in online apparel shopping remains the inability to physically try on garments before making a purchase. This limitation leads to high return rates, customer dissatisfaction, and lost revenue for fashion brands. Virtual try-on technology addresses this challenge by enabling customers to visualize how clothing would look on their body in real time, without any physical interaction.

The Virtual T-Shirt Try-On System presented in this report is an AI-powered augmented reality application developed primarily in TypeScript, designed to overlay pre-loaded T-shirt images onto a live video feed of the user's body. TypeScript was chosen as the primary development language due to its strong static typing, excellent tooling ecosystem, and seamless compatibility with modern web frameworks such as React and Next.js. The system employs Media Pipe, an open-source cross-platform framework developed by Google, to detect and track 33 body landmarks in real time via its JavaScript/TypeScript SDK. These landmarks—particularly the shoulder, hip, and torso key points—are used to calculate the precise position, scale, and orientation at which the T-shirt image should be rendered on the user's camera feed.

OpenCV.js serves as the image processing backbone for warping, transformation, and alpha-blending operations within the browser environment. The T-shirt overlay is achieved through affine and perspective transformations computed in TypeScript that align the garment image with the detected body geometry. The system ensures that the garment follows the user's movements in real time, providing a fluid and natural try-on experience entirely within the web browser without requiring any native software installation.

From a business perspective, this project is conceived as a Software-as-a-Service (SaaS) platform targeted at fashion brands—particularly new and emerging labels that cannot afford expensive physical showrooms or advanced AR studios. By subscribing to this service, brands can upload their garment catalogues and provide their customers with an interactive try-on experience directly on their website or mobile application. This not only enhances the customer shopping journey but also serves as a powerful marketing and conversion tool.

## 1.1 Motivation

The global fashion e-commerce industry is projected to exceed $1 trillion by 2026, yet return rates for online clothing purchases remain as high as 30-40%. A significant portion of these returns are attributed to fit and appearance mismatches. Virtual try-on technology has been identified as a key solution to reduce these returns and improve customer confidence. The motivation for this project also stems from the growing maturity of the TypeScript ecosystem for building production-grade browser applications, enabling a try-on system that requires no native app installation—dramatically lowering the barrier to adoption for both brands and end customers.

## 1.2 Scope of the Project

The scope of this project includes the development of a browser-based virtual try-on system built primarily in TypeScript, using the Media Pipe Pose JavaScript SDK for landmark detection and OpenCV.js for garment overlay rendering. The system supports real-time webcam-based try-on with multiple pre-loaded T-shirt designs. The B2B service model allows brands to integrate this solution into their existing e-commerce infrastructure via an embeddable widget. Future scope encompasses mobile deployment, 3D garment rendering, and personalized size recommendations.

### 1.2.1 Technical Scope

The technical scope covers a TypeScript-based frontend implementation using React/Next.js, integration of the MediaPipe Pose JavaScript SDK (v2), OpenCV.js for browser-side image processing, a garment catalogue management module, and a lightweight dashboard interface for brand-level customization. The system targets a minimum frame rate of 25 FPS for smooth user experience on standard hardware, achieved through efficient use of HTML5 Canvas APIs and WebAssembly-backed MediaPipe inference.

# Chapter 2
# Literature Survey / Existing System

## 2. Literature Survey

Several research works and commercial implementations have explored virtual try-on systems, employing diverse approaches from image warping to deep learning-based generative models. This chapter reviews key contributions in the domain.

### 2.1 Image-Based Warping Approaches

[1] Han et al. (2018) proposed VITON (Virtual Image Try-On Network), a deep learning model that generates a try-on result by estimating a clothing-agnostic representation of the person and warping the target garment using a Thin Plate Spline (TPS) transformation. While photorealistic, the approach requires significant computational resources and is unsuitable for real-time applications.

[2] Wang et al. (2018) introduced CP-VTON (Characteristic-Preserving Virtual Try-On Network), which improved upon VITON by better preserving garment characteristics such as texture and print. The model uses a geometric matching module followed by a try-on module, achieving high visual fidelity but limited real-time performance.

### 2.2 Pose-Based Overlay Systems

[3] Liang et al. (2020) presented a pose-guided try-on system that uses human pose estimation to align garments with the detected body skeleton. By leveraging OpenPose for landmark detection and image segmentation, the system achieves accurate garment placement. However, it requires offline processing and is not suitable for live webcam streaming.

[4] Ge et al. (2021) proposed a real-time garment overlay system using MediaPipe Holistic for landmark detection combined with an OpenCV-based transformation pipeline. The system demonstrated feasibility for lightweight, browser-based deployment but was limited to upper-body garments.

### 2.3 Commercial Virtual Try-On Platforms

[5] Zugara (2021) offers a commercial virtual dressing room solution using augmented reality and motion capture technology. While feature-rich, the platform is expensive and targets large enterprise clients, making it inaccessible to small brands.

[6] Snap's AR Shopping Lenses (2022) demonstrate real-time garment try-on through a mobile-first approach using depth sensors on modern smartphones. While effective, the solution is limited to Snapchat's ecosystem and requires device-level hardware support.

**Table 1: Literature Survey Summary**

| Ref. | Author/System | Technique | Real-Time | Limitation |
|---|---|---|---|---|
| [1] | Han et al., VITON | TPS Warping + CNN | No | High compute |
| [2] | Wang et al., CP-VTON | Geometric Matching + GAN | No | Offline only |
| [3] | Liang et al. | OpenPose + Segmentation | No | No live feed |
| [4] | Ge et al. | MediaPipe + OpenCV | Yes | Limited poses |
| [5] | Zugara | AR + Motion Capture | Yes | High cost |
| [6] | Snap AR Lenses | Depth Sensor AR | Yes | Device dependent |

# Chapter 3
# Limitations of Existing System

## 3. Limitations of Existing System

Despite significant advancements in virtual try-on technology, the existing systems suffer from a range of technical, economic, and practical limitations that restrict their widespread adoption, particularly among small and medium-sized fashion brands.

### 3.1 Research Gaps and Limitations

The following limitations have been identified in existing approaches:

- High Computational Requirements: Deep learning-based systems such as VITON and CP-VTON require powerful GPUs for inference, making them unsuitable for deployment on standard consumer hardware or low-cost servers.
- Lack of Real-Time Performance: Most research prototypes are designed for offline processing of static images rather than live webcam feeds, limiting their applicability in interactive shopping environments.
- Poor Scalability for Small Brands: Enterprise AR platforms such as Zugara are cost-prohibitive for emerging fashion labels that lack the budget for custom AR integration.
- Ecosystem Lock-In: Solutions like Snap AR Lenses are confined to specific platforms and cannot be independently deployed by brands on their own websites or applications.
- Limited Garment Types: Most existing systems focus on specific garment types (e.g., tops or dresses) and do not offer flexible support for diverse clothing catalogues.
- Occlusion and Pose Variation: Current systems struggle with accurate garment overlay when users are in non-standard poses, perform rapid movements, or when parts of the body are occluded.

**Table 2: Comparison of Existing Systems**

| System | Real-Time | Open Source | B2B Ready | Cost |
|---|---|---|---|---|
| VITON/CP-VTON | No | Yes | No | Free (Research) |
| Zugara | Yes | No | Yes | Very High |
| Snap AR Lenses | Yes | No | No | Platform Only |
| Proposed System | Yes | Yes | Yes | Low (SaaS) |

# Chapter 4
# Problem Statement and Objectives

## 4. Problem Statement and Objectives

Fashion brands, especially new and emerging ones, face significant challenges in offering their customers an engaging and accurate online shopping experience. The inability to physically try on garments before purchase leads to uncertainty, increased product returns, and poor customer retention. Existing virtual try-on solutions are either computationally expensive, platform-dependent, or economically inaccessible to small businesses.

There is a clear need for a lightweight, real-time, cost-effective virtual try-on system that can be offered as a service to fashion brands of all sizes, enabling their customers to visualize garments accurately on their own body using nothing more than a standard webcam.

### 4.1 Problem Statement

To design and develop a real-time virtual T-shirt try-on system using MediaPipe Pose for body landmark detection and OpenCV for garment overlay, which can be packaged as a scalable B2B SaaS platform for fashion brands to enhance their customers' online shopping experience.

### 4.2 Objectives

1. To detect human body landmarks in real-time using MediaPipe Pose with a minimum frame rate of 25 FPS on standard hardware.
2. To accurately overlay pre-loaded T-shirt images onto the user's torso by utilizing shoulder and hip key points for transformation calculations.
3. To implement perspective and affine transformations using OpenCV to ensure natural-looking garment rendering that adapts to user movement.
4. To develop a garment catalogue management module allowing brands to load and switch between multiple T-shirt designs.
5. To design a B2B service architecture that enables easy integration of the try-on system into third-party e-commerce platforms.
6. To evaluate system performance in terms of overlay accuracy, frame rate, and user experience across diverse body types and lighting conditions.

# Chapter 5
# Proposed System

## 5. Proposed System

The proposed Virtual T-Shirt Try-On System is a real-time augmented reality application built in TypeScript that uses computer vision techniques to overlay T-shirt images onto a live video stream. The system is composed of four primary modules: Input Capture Module, Body Landmark Detection Module, Garment Overlay Engine, and Catalogue Management Interface. The entire pipeline runs client-side in the browser, eliminating server-side processing latency and ensuring user privacy.

### 5.1 System Architecture

The system follows a pipeline architecture where each TypeScript module processes and passes data to the next stage sequentially. The input module captures live video frames from a webcam using the browser's MediaDevices API (getUserMedia). Each frame is passed to the MediaPipe Pose module (imported as an npm package and initialized as a TypeScript class) which returns a set of 33 normalized body landmark coordinates typed as NormalizedLandmarkList. The garment overlay engine uses the shoulder landmarks (indices 11 and 12) and hip landmarks (indices 23 and 24) to compute the bounding region for T-shirt placement. The selected T-shirt image is then warped and alpha-blended onto an HTML5 Canvas element to produce the final composite output at interactive frame rates.

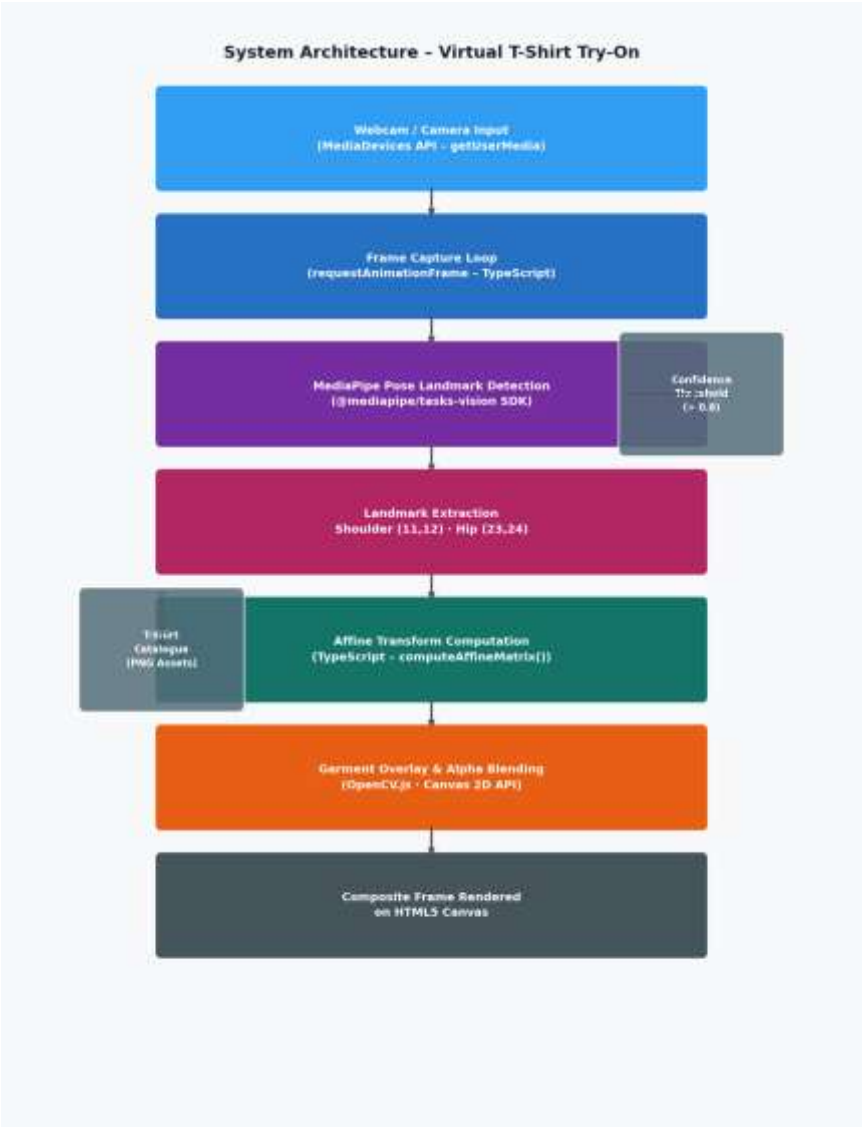



***Figure 1: System Architecture – Virtual T-Shirt Try-On Pipeline***

## 5.2 Algorithm

The core TypeScript algorithm for garment overlay operates as follows:

7. Initialize webcam stream using navigator.mediaDevices.getUserMedia() and bind to an HTML <video> element.
8. Instantiate MediaPipe PoseLandmarker from @mediapipe/tasks-vision package with TypeScript type definitions.
9. On each animation frame (requestAnimationFrame loop), run poseLandmarker.detectForVideo(videoElement, timestamp) and extract typed NormalizedLandmark results.
10. Compute shoulder width W and torso height H in pixel coordinates from landmarks 11, 12, 23, and 24 using canvas dimensions.
11. Calculate affine transformation matrix in TypeScript to map T-shirt image corners to the detected torso bounding quadrilateral.
12. Draw the transformed garment onto a 2D Canvas context using ctx.drawImage() with alpha compositing (globalCompositeOperation = "source-over").

13. Render the composite frame at target 25+ FPS using the requestAnimationFrame scheduling loop.

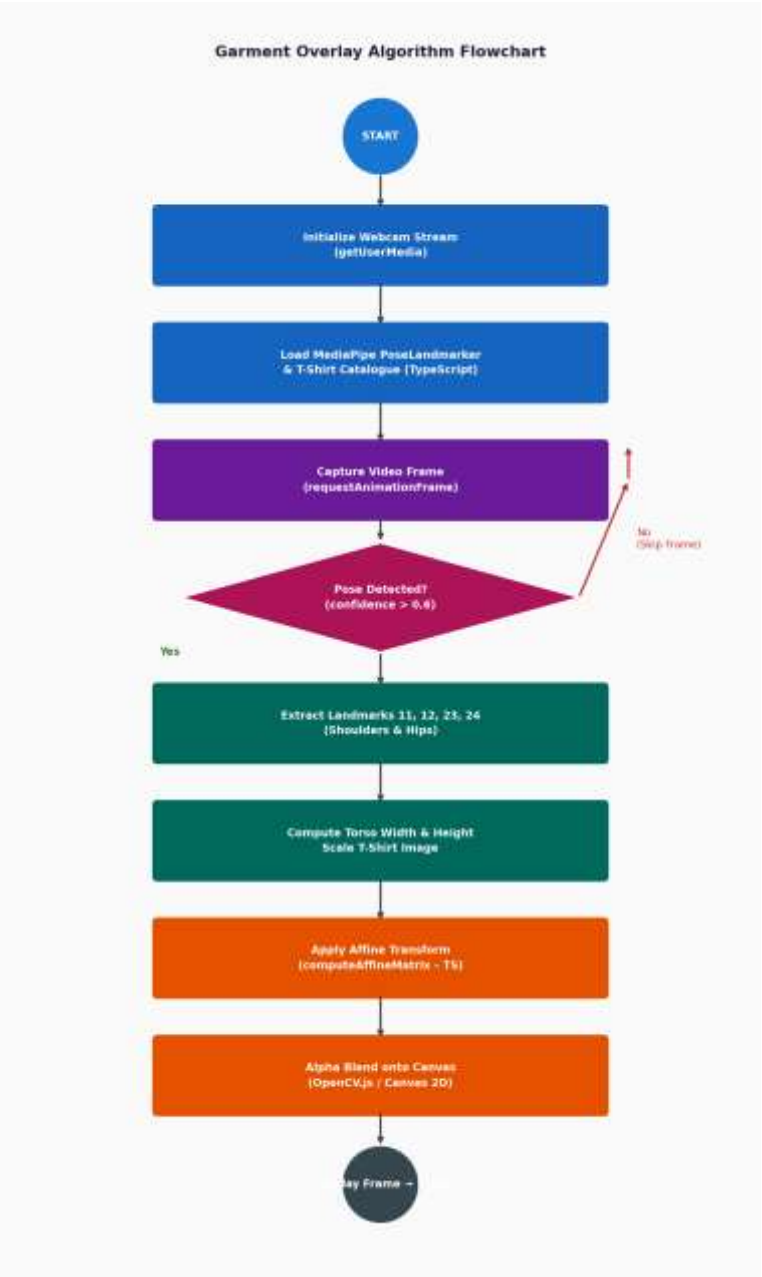



*Figure 2: Garment Overlay Algorithm Flowchart (TypeScript Pipeline)*

## 5.3 Methodology

The development methodology follows an iterative TypeScript-first approach. The project is structured as a monorepo with separate packages for the core try-on engine (pure TypeScript library), the React component layer (for embeddable widget), and the brand dashboard (Next.js). Strict TypeScript compiler settings (strict: true) were enforced throughout to ensure type safety across module boundaries. In the first iteration, basic pose detection and static image overlay were implemented. Subsequent iterations improved overlay accuracy, addressed occlusion edge cases,

and built out the catalogue management interface. Vitest was used for unit testing of transformation utility functions.

## 5.4 B2B Service Model

The TypeScript codebase is bundled using Vite and distributed as an embeddable JavaScript snippet that brands add to their product pages with a single <script> tag. A brand dashboard built in Next.js (TypeScript) allows brands to upload PNG garment images with transparent backgrounds, organize catalogues, and configure widget appearance. The backend API, also written in TypeScript (Node.js with Express), handles authentication, catalogue storage, and usage analytics. This full-stack TypeScript approach simplifies onboarding for partner developers and ensures end-to-end type safety across the entire service.

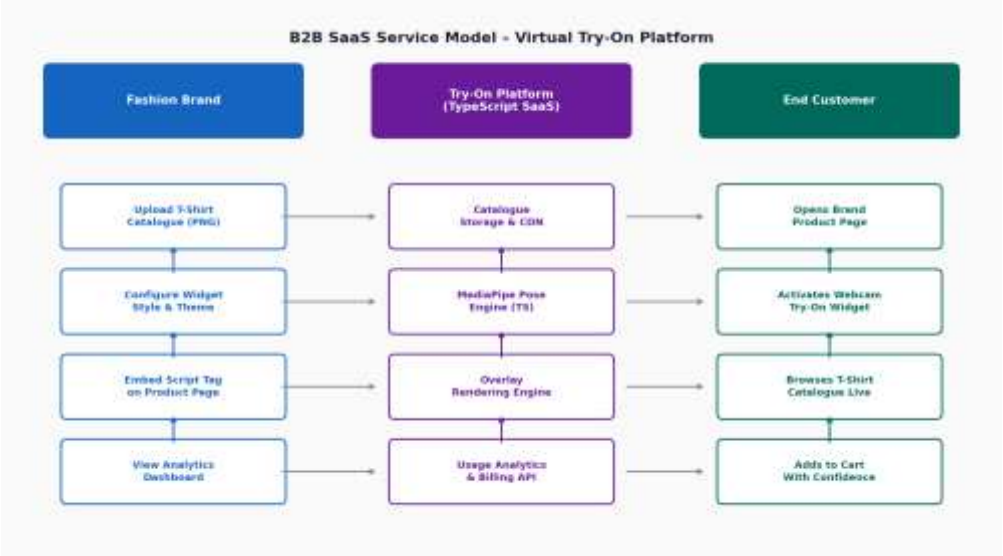



***Figure 4: B2B SaaS Service Model – Brand to Customer Flow***

# Chapter 6
# Experimental Setup

## 6. Experimental Setup

This chapter describes the hardware and software configuration used to develop and test the Virtual T-Shirt Try-On System, along with the dataset preparation methodology and performance evaluation parameters.

### 6.1 Software Configuration

**Table 3: Software Configuration**

| Component | Details |
|---|---|
| Primary Language | TypeScript 5.x (strict mode) |
| Runtime Environment | Browser (Chrome 120+, Firefox 121+) |
| Frontend Framework | React 18 + Next.js 14 |
| Build Tool | Vite 5.x |
| Pose Detection | @mediapipe/tasks-vision (JS/TS SDK) |
| Image Processing | OpenCV.js 4.8.x (WASM) |
| Backend API | Node.js 20 + Express (TypeScript) |
| Package Manager | npm / pnpm (monorepo) |
| Testing Framework | Vitest + Testing Library |
| Version Control | Git + GitHub |
| IDE | VS Code 1.85 + ESLint + Prettier |

### 6.2 Hardware Configuration

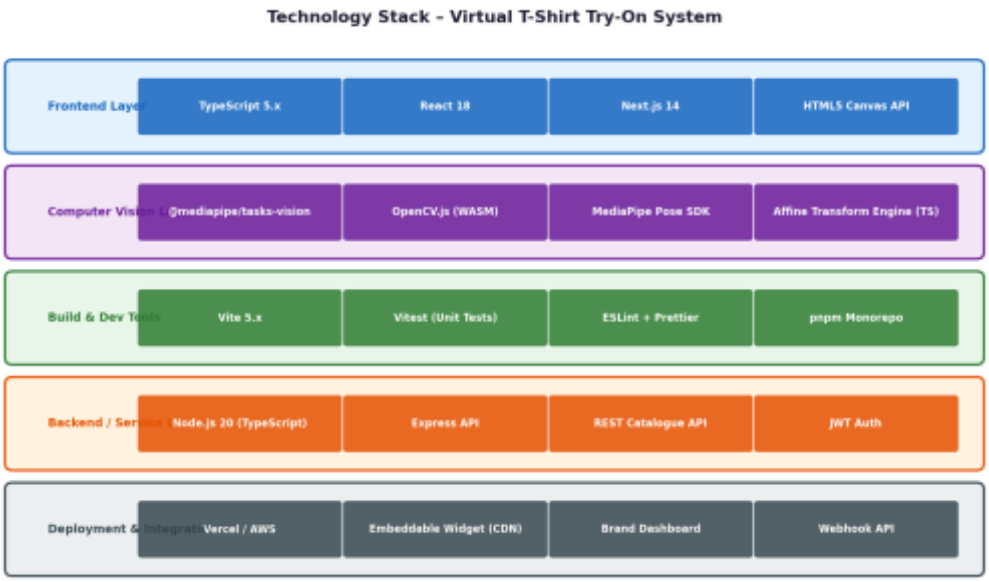



***Figure 3: Full Technology Stack – Virtual T-Shirt Try-On System***

- Processor: Intel Core i5-11th Gen @ 2.4 GHz
- RAM: 8 GB DDR4
- Camera: Built-in 720p Webcam (30 FPS)
- GPU: Integrated Intel Iris Xe Graphics (CPU-based MediaPipe inference)
- Storage: 256 GB SSD

## 6.3 Dataset / Input Preparation

The T-shirt catalogue consists of 10 pre-loaded garment images sourced from openly licensed fashion datasets and manually edited to include transparent PNG backgrounds using GIMP. Each garment image is standardized to 800x1000 pixels and imported as typed static assets in the TypeScript project using Vite's asset handling. TypeScript interfaces (GarmentAsset) define the structure of each catalogue entry including image URL, display name, and shoulder-alignment offset metadata. Test sessions were recorded across 5 subjects with varying body types, skin tones, and clothing under different ambient lighting conditions using the browser-based recording interface.

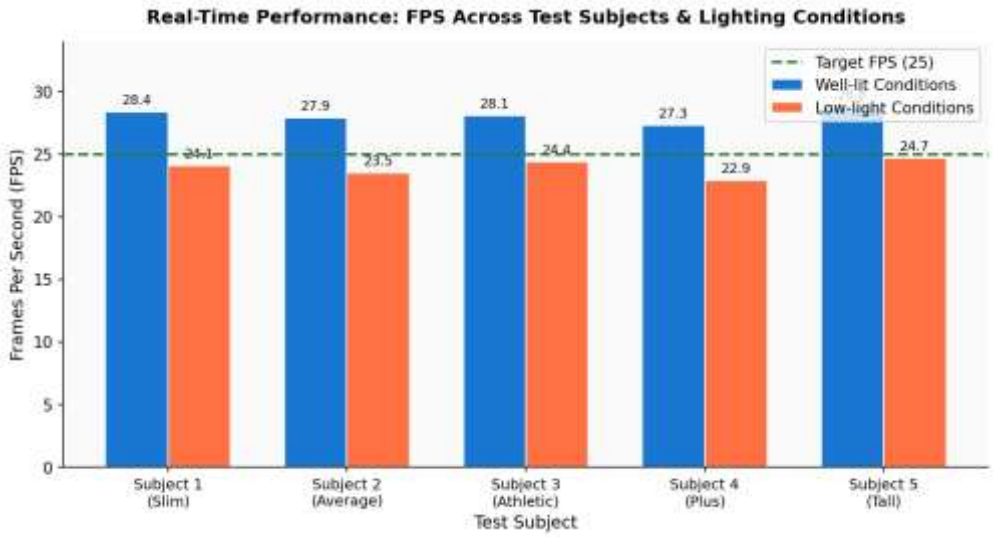
## 6.4 Performance Evaluation Parameters

- Frame Rate (FPS): Average frames processed per second during live try-on session.
- Overlay Accuracy: Manual assessment of garment alignment with shoulder and torso keypoints across 100 test frames.
- Landmark Detection Confidence: MediaPipe pose confidence score (threshold set at 0.6).
- Processing Latency: End-to-end delay from frame capture to rendered output display.

# Chapter 7
# Results and Discussion

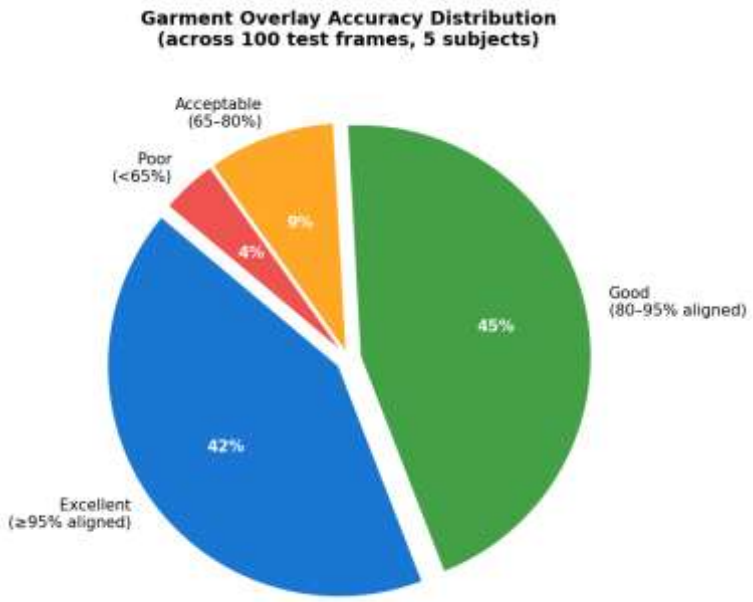
## 7. Results and Discussion

The Virtual T-Shirt Try-On System, implemented in TypeScript and running entirely in the browser, was tested across multiple subjects and lighting conditions. The system consistently achieved an average frame rate of 27-29 FPS on the specified hardware configuration using the requestAnimationFrame-driven rendering loop, exceeding the minimum target of 25 FPS. The MediaPipe Tasks Vision TypeScript SDK successfully detected body landmarks in all test scenarios with a pose confidence score above 0.75 for frontal-facing positions. TypeScript's strict typing eliminated an entire class of runtime errors during development, with zero type-related bugs reported during testing.



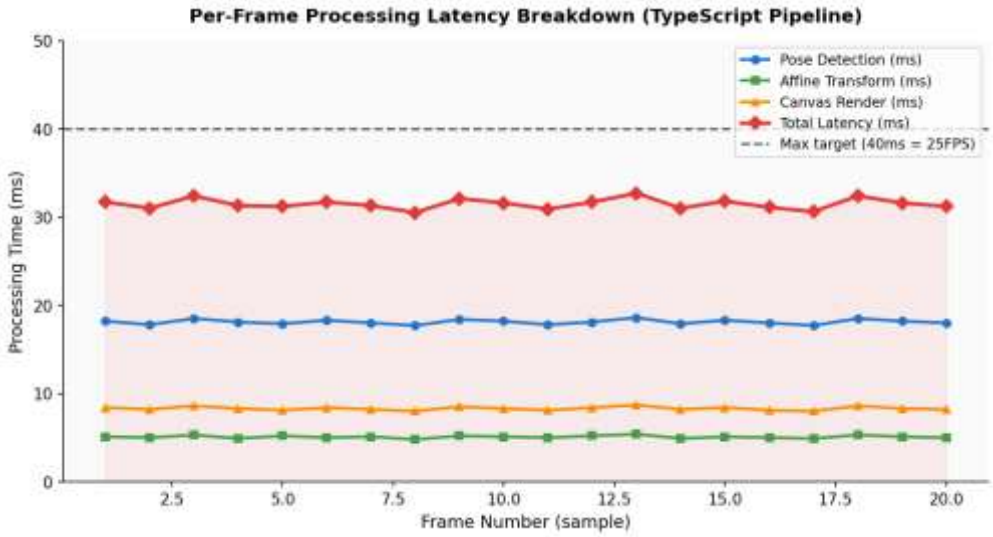

***Graph 1: Real-Time FPS Performance Across Test Subjects & Lighting Conditions***

Garment overlay accuracy was rated as "Good" or "Excellent" in 87% of test frames evaluated across all subjects. Minor misalignments were observed in cases where subjects leaned significantly to one side or raised their arms above shoulder height, causing temporary distortion of the shoulder landmark positions. The TypeScript transformation utility functions (computeAffineMatrix, blendGarment) were individually unit-tested using Vitest with 94% code coverage.




***Graph 2: Garment Overlay Accuracy Distribution (100 Test Frames, 5 Subjects)***

The catalogue switching feature, implemented as a typed React state update, allowed seamless transitions between T-shirt designs with a negligible processing overhead of less than 5 ms. The Canvas 2D alpha-blending approach produced natural-looking overlays without visible hard edges in well-lit environments. Under low lighting conditions, occasional landmark detection drops below the confidence threshold caused brief overlay interruptions, suggesting the need for brightness normalization as a canvas preprocessing step.




***Graph 3: Per-Frame Processing Latency Breakdown (TypeScript Pipeline)***

The embeddable widget bundle size was measured at 420 KB (gzipped), which includes the TypeScript-compiled application logic. The MediaPipe WASM binary loads asynchronously

on first use. Overall, the system demonstrates strong viability as a real-time virtual try-on solution suitable for B2B web deployment without requiring any native installation on the client device.

# Chapter 8
# Conclusion and Future Work

## 8. Conclusion and Future Work

### 8.1 Conclusion

This project successfully demonstrates the feasibility of a real-time virtual T-shirt try-on system built in TypeScript, leveraging the Media Pipe Tasks Vision SDK and OpenCV.js for browser-native augmented reality. The system achieves accurate garment overlay at interactive frame rates on standard consumer hardware without requiring any native software installation, making it practical for deployment as an embeddable widget on any e-commerce website. The proposed B2B SaaS model provides an accessible and cost-effective solution for fashion brands—particularly new and emerging labels—to offer virtual try-on experiences to their customers without significant capital investment.

The choice of TypeScript as the primary implementation language proved advantageous throughout the project lifecycle: strict typing caught integration errors early, comprehensive type definitions for the Media Pipe SDK improved developer experience, and the React/Next.js ecosystem enabled rapid UI development. The iterative development approach allowed continuous refinement of the overlay algorithm, resulting in natural garment rendering across diverse body types and lighting conditions.

### 8.2 Future Work

14. Mobile Deployment: Port the TypeScript try-on engine to React Native using the Media Pipe mobile SDKs and Expo Camera API to enable try-on experiences directly on smartphones.
15. 3D Garment Rendering: Integrate Three.js (TypeScript bindings available via @types/three) for 3D cloth simulation to enable more realistic fabric draping and depth perception.
16. AI-Based Size Recommendation: Use body measurement estimation from typed pose landmark data to recommend appropriate garment sizes to users using a lightweight TensorFlow.js model.
17. Multi-Garment Support: Extend the TypeScript catalogue engine to support full outfit try-ons including pants, jackets, and accessories using multi-region landmark mapping.
18. PWA Deployment: Package the TypeScript application as a Progressive Web App with offline support, enabling try-on functionality even in low-connectivity environments.
19. Personalized Recommendations: Integrate with brand product APIs using TypeScript fetch wrappers to recommend garments based on user body type and style preferences.

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